

3-4 General Expression for Gravitational Potential Energy



Source / 來源

- 選修物理(二) 課堂講義 (4624R51) 教用版，第3章 功與能量，§ 3-4 重力位能的一般表示式，pp. 134–140
- Halliday, Resnick & Walker, *Fundamentals of Physics*, Ch 13: Gravitation
- Young & Freedman, *University Physics with Modern Physics*, Ch 13: Gravitation

Examples

Example 1 — Choice of Zero for Gravitational Potential (*p. 136*)

The gravitational force between a mass m and Earth of mass M varies with center-to-center distance R as $F = GMm/R^2$. Taking infinity as the zero of potential, find:

- (a) The potential energy at the Earth's surface R_E .
- (b) The potential energy at the surface if the zero is instead chosen at $2R_E$.



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Solution

Try It 1 — Shifted Zero Level (*p. 136*)

With $U(\infty) = 0$, an object has $U = -3GMm/(4R)$. If the Earth's surface is redefined as the zero level, what is the object's new potential energy?

(A) $-GMm/(3R)$ (B) $-GMm/(2R)$ (C) GMm/R (D) $GMm/(2R)$ (E) $GMm/(4R)$

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 Solution

Example 2 — Potential of a Three-Body System (*p. 137*)

Three equal masses m form an equilateral triangle of side L . Find the total gravitational potential energy. If one mass is moved to infinity while the other two stay fixed, find ΔU .

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 Solution

Try It 2 — Four Masses at a Square (*p. 137*)

Four equal masses M occupy the vertices of a square of side L . One mass is moved to infinity while the other three remain fixed. Find the increase in system potential energy.

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Quick Check

General Gravitational Potential

Q1. A satellite moves in an elliptical orbit with nearest distance R and farthest distance $2R$ from Earth's center. What work does gravity do as it moves from nearest to farthest point?

(A) $2GMm/R$ (B) $-GMm/(2R)$ (C) $GMm/(2R)$ (D) $-GMm/R$ (E) GMm/R

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Q2. Earth has mass M and radius R . A spacecraft has mass m . Which statements are correct? (Select 3.)

(A) With infinity as zero, $U(R) = +GMm/R$.

(B) With infinity as zero, $U(2R) = -GMm/(2R)$.

(C) With infinity as zero, raising the craft from R to $2R$ changes its potential by $-GMm/(2R)$.

(D) With the surface as zero, $U(2R) = +GMm/(2R)$.

(E) The potential change from R to $2R$ is $+GMm/(2R)$ regardless of zero choice.

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 Solution

Q3. An asteroid of mass 2.0×10^{15} kg falls from rest at infinity onto Earth ($M = 6.0 \times 10^{24}$ kg, $R = 6.4 \times 10^6$ m, $G = 6.7 \times 10^{-11}$ SI). The energy released is equivalent to how many 6.0×10^{13} J atomic bombs?

(A) 2×10^3 (B) 2×10^4 (C) 2×10^7 (D) 2×10^8 (E) 2×10^9

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 Solution

Q4. Two stars of masses m and $2m$ are separated by r . If their separation becomes $4r$, what is the change in gravitational potential energy?

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💡 Solution

Q5. A uniform ring of mass M and radius R surrounds a point mass m at its center. What is the potential-energy change in moving m from infinity to the center?

(A) GMm/R (B) $GMm/(2R)$ (C) 0 (D) $-GMm/R$ (E) $-GMm/(2R)$

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💡 Solution

Q6. Three equal point masses m occupy the vertices of a right triangle with sides $3a, 4a, 5a$. The right-angle mass is moved slowly to infinity. What work does gravity do?

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💡 Solution

Challenge

General Gravitational Potential Energy

Q1. A satellite's circular orbit has period 180 min and potential energy U (infinity is zero). What is its potential energy in a geosynchronous orbit (period 24 h)?

(A) $U/2$ (B) $U/3$ (C) $U/4$ (D) $U/5$ (E) $U/6$

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 Solution

Q2. A ring of mass M and radius R lies horizontally. A point mass m is at height R above its center. What is their gravitational potential energy?

(A) $-GMm/(2R)$ (B) $-2GMm/R$ (C) $-GMm/R$ (D) $-GMm/(2R)$ (E) $-2GMm/R$

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 Solution

Square System

Q3. Four equal masses m are at the corners of a square of side a . Find the total potential energy.

(A) $-Gm^2/a$ (B) $-4Gm^2/a$ (C) $-6Gm^2/a$ (D) $-(4 + \sqrt{2})Gm^2/a$ (E) 0

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 Solution

Q4. If any two masses are removed to infinity from the square, what is the minimum external work?

(A) Gm^2/a (B) $4Gm^2/a$ (C) $(3 + \sqrt{2})Gm^2/a$ (D) $(4 + 1/\sqrt{2})Gm^2/a$ (E) $(4 + \sqrt{2})Gm^2/a$

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 Solution

Q5. A satellite of mass m circles a moon of mass M with kinetic energy K . With infinity as zero potential, what is its gravitational potential energy?

 Solution

Q6. Near Earth's surface a ball reaches height h for a given launch speed. Using the exact potential $U = -GMm/r$ instead of mgy , what maximum height does it reach? (Earth radius R .)

 Solution

Exam Archive

Q1 — Circular-Orbit Energy of a Four-Mass Square (*p. 140*)

Four equal masses m occupy the vertices of a square of side a , with infinity as zero. What is the total potential energy?

 Solution

Q2 — Removing Two Masses (*p. 140*)

From the same square, two masses are moved one after another to infinity. What minimum external work is required?

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 Solution

Q3 — Circular Satellite and Potential Energy (*p. 140*)

A satellite of kinetic energy K moves in a circular orbit around a moon. Find its gravitational potential energy with infinity as zero.

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 Solution

Q4 — Exact Height of a Projectile (*p. 140*)

A projectile reaches height h in the uniform-field approximation. What height does it reach when the inverse-square variation of gravity is included?

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 Solution

Glossary

English	中文	Symbol / Unit
gravitational potential energy	重力位能	$U_G = -GMm/r$ [J]
universal gravitation	萬有引力	$F = GMm/r^2$
gravitational field	重力場	—
inverse-square law	平方反比定律	$F \propto 1/r^2$
reference (zero) level	零位面	$U = 0$
potential difference	位能差	ΔU
point mass	質點	—
center-to-center distance	質心距離	r
system potential energy	系統位能	$U = \sum_{i < j} -Gm_i m_j / r_{ij}$
ring	圓環	—
equilateral triangle	正三角形	
square	正方形	
Kepler's third law	克卜勒行星運動第三定律	$T^2 \propto r^3$